

Orchestration Report

Status: 3 completed, 1 failed

Session: 6a3f3710d163a13dc7ec9030

Generated: 2026-06-27T03:11:52.065Z

Overview

Metric	Value
Total Tasks	4
Completed	3
Failed	1
Pending	0
Tool Calls	133
Errors	2
Duration	10m 41s

Agent Breakdown

Agent	Tasks
orchestrator	59
deployer	46
tuner	28

Summary

Status: 3 completed, 1 failed

Duration: 10m 41s

Tasks: 3 completed, 1 failed, 0 pending

Tool calls: 133 (2 tool errors)

- Errors by tool: mongodb_find(2)

Findings

Results (3)

[orchestrator] Execute skill: EMULATOR_SPEC

Still pending. Let me give the agents more time to pick up and process these tasks.

Used 58 tool call(s): list_memory_keys, mongodb_list_collections, mongodb_find, mongodb_find, mongodb_find, mongodb_find, mongodb_find, mongodb_find, mongodb_find, mongodb_find, read_memory, read_memory, mongodb_find, mongodb_find, mongodb_find, mongodb_find, write_task, write_task, write_task, write_memory, list_tasks, list_tasks, read_task_result, read task result, read task result, read task result

Structured details:

```
{
  "stopReason": "max_tool_rounds",
  "toolCalls": [
    {
      "tool": "list_memory_keys"
    },
    {
      "tool": "mongodb_list_collections",
      "input": {
        "includeStats": true
      }
    },
    {
      "tool": "mongodb_find",
      "input": {
        "collection": "convergence_criteria",
        "limit": 5
      }
    },
    {
      "tool": "mongodb_find",
      "input": {
        "collection": "convergence_state",
        "limit": 5,
        "sort": "{\"_id\": -1}"
      }
    },
    {
      "tool": "mongodb_find",
      "input": {
        "collection": "sentinel_state",
        "limit": 3,
        "sort": "{\"_id\": -1}"
      }
    },
    {
      "tool": "mongodb_find",
      "input": {
        "collection": "deployer_state",
        "limit": 3,
        "sort": "{\"_id\": -1}"
      }
    },
    {
      "tool": "mongodb_find",
      "input": {
        "collection": "researcher_state",
        "limit": 3,
        "sort": "{\"_id\": -1}"
      }
    }
  ]
}
```

```
    },
    {
      "tool": "mongodb_find",
      "input": {
        "collection": "convergence_metrics",
        "limit": 5,
        "sort": "{\"_id\": -1}"
      }
    },
    {
      "tool": "mongodb_find",
      "input": {
        "collection": "researcher_state",
        "filter": "{\"procedure\": \"insignia_subnet_tuner\"}",
        "limit": 3,
        "sort": "{\"_id\": -1}"
      }
    },
    {
      "tool": "mongodb_find",
      "input": {
        "collection": "convergence_metrics",
        "filter": "{\"procedure\": \"insignia_subnet_tuner\"}",
        "limit": 5,
        "sort": "{\"_id\": -1}"
      }
    },
    {
      "tool": "mongodb_find",
      "input": {
        "collection": "experiment_results",
        "filter": "{\"procedure\": \"insignia_subnet_tuner\"}",
        "limit": 3,
        "sort": "{\"_id\": -1}"
      }
    },
    {
      "tool": "mongodb_find",
      "input": {
        "collection": "tuner_optimization",
        "filter": "{\"procedure\": \"insignia_subnet_tuner\"}",
        "limit": 3,
        "sort": "{\"_id\": -1}"
      }
    },
    {
      "tool": "read_memory",
      "input": {
        "key": "emulator_execution_status"
      }
    },
  ],
```

```

{
  "tool": "read_memory",
  "input": {
    "key": "emulator_loop_iteration_plan"
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "simulation_runs",
    "detail_level": "L1",
    "limit": 5,
    "sort": "{\"_id\": -1}"
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "experiment_results",
    "detail_level": "L1",
    "limit": 5,
    "sort": "{\"_id\": -1}"
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "breach_alerts",
    "detail_level": "L1",
    "limit": 3,
    "sort": "{\"_id\": -1}"
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "defense_strategies",
    "detail_level": "L1",
    "limit": 10,
    "sort": "{\"_id\": -1}"
  }
},
{
  "tool": "write_task",
  "input": {
    "_raw": "{\"title\": \"Deployer: Verify chain, fund wallets, deploy monitoring + PC-VH-0"
  }
},
{
  "tool": "write_task",
  "input": {

```

```

    "title": "Tuner: Initialize NSGA-II v5 gen0 population seeded with V13-R3 knee point",
    "description": "You are the tuner agent in the Insignia subnet emulator loop. The system",
    "assignedTo": "tuner",
    "priority": 9,
    "input": {
      "n_var": 41,
      "n_obj": 4,
      "population_size": 30,
      "seed_config": "V13-R3-KP-020-a3c7",
      "objectives": [
        "neg_mean_honest_score",
        "attack_breach_rate",
        "score_variance",
        "neg_score_separation"
      ],
      "algorithm": "NSGA-II",
      "repair": "weight_normalization_L1_L2",
      "procedure": "insignia_subnet_tuner"
    }
  },
  {
    "tool": "write_task",
    "input": {
      "assignedTo": "deployer",
      "description": "You are the deployer agent in the Insignia subnet emulator loop. The sys
...(truncated)

```

[tuner] Tuner: Initialize NSGA-II v5 gen0 population seeded with V13-R3 knee point

The Gen0 population has been generated successfully. Now I'll store the population in MongoDB and

Structured details:

```

```json
{
 "toolCallCount": 28,
 "toolCalls": [
 {
 "tool": "read_memory",
 "input": {
 "key": "tuner_state"
 }
 },
 {
 "tool": "mongodb_find_one",
 "input": {
 "collection": "tuner_optimization",
 "filter": "{\"status\": {\"$in\": [\"initialized_awaiting_evaluation\", \"completed\"]}}"
 }
 }
],
 {

```

```

 "tool": "mongodb_aggregate",
 "input": {
 "collection": "convergence_state",
 "pipeline": "[{\"$match\": {\"procedure\": \"insignia_subnet_tuner\"}}, {\"$sort\": {\"u
 }
 },
 {
 "tool": "mongodb_aggregate",
 "input": {
 "collection": "tuner_optimization",
 "pipeline": "[{\"$match\": {\"config_id\": \"V13-R3-KP-020-a3c7\"}}, {\"$limit\": 1}]"
 }
 },
 {
 "tool": "mongodb_list_collections",
 "input": {
 "includeStats": true
 }
 },
 {
 "tool": "mongodb_find_one",
 "input": {
 "collection": "convergence_state",
 "filter": "{}"
 }
 },
 {
 "tool": "mongodb_aggregate",
 "input": {
 "collection": "tuner_optimization",
 "pipeline": "[{\"$match\": {\"config_id\": {\"$options\": \"i\", \"$regex\": \"V13.*KP\"
 }
 },
 {
 "tool": "mongodb_aggregate",
 "input": {
 "collection": "experiment_results",
 "pipeline": "[{\"$match\": {\"config_id\": {\"$options\": \"i\", \"$regex\": \"V13.*KP\"
 }
 },
 {
 "tool": "mongodb_aggregate",
 "input": {
 "collection": "experiment_results",
 "pipeline": "[{\"$sort\": {\"timestamp\": -1}}, {\"$limit\": 10}]"
 }
 },
 {
 "tool": "mongodb_find",
 "input": {
 "collection": "experiment_results",

```

```

 "filter": "{}",
 "limit": 5,
 "detail_level": "L1"
 }
},
{
 "tool": "mongodb_aggregate",
 "input": {
 "collection": "tuner_optimization",
 "pipeline": "[{\"$sort\": {\"timestamp\": -1}}, {\"$limit\": 10}]"
 }
},
{
 "tool": "mongodb_aggregate",
 "input": {
 "collection": "simulation_results",
 "pipeline": "[{\"$sort\": {\"timestamp\": -1}}, {\"$limit\": 5}]"
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "cat > /tmp/gen0_population.py << 'EOF'\n#!/usr/bin/env python3\n\"\\\"\\\"\\nGen0
...(truncated)

```

#### [deployer] Deployer: Verify chain, fund wallets, deploy monitoring stack, deploy PC-VH-006

Good - the metrics endpoint is running but most values are 0 (not yet populated with current sta

Structured details:

```
```json
```

```

{
  "toolCallCount": 46,
  "toolCalls": [
    {
      "tool": "read_memory",
      "input": {
        "key": "deployer_state"
      }
    },
    {
      "tool": "read_memory",
      "input": {
        "key": "tuner_state"
      }
    },
    {
      "tool": "read_memory",
      "input": {
        "key": "sentinel_state"
      }
    }
  ]
}

```

```
    },
    {
      "tool": "shell_execute",
      "input": {
        "command": "find /mnt -maxdepth 1 -type d 2>/dev/null | head -20; ls -la /mnt/Synth2/ 2>/dev/null",
        "description": "Check NAS mount points and project directories"
      }
    },
    {
      "tool": "shell_execute",
      "input": {
        "command": "find . -maxdepth 4 -name \"docker-compose*.yaml\" -o -name \"check_chain*\" -type f 2>/dev/null",
        "description": "Find relevant scripts and config files"
      }
    },
    {
      "tool": "shell_execute",
      "input": {
        "command": "find . -maxdepth 4 -type d -name \"testnet\" 2>/dev/null; find . -maxdepth 4 -type f -name \"*.conf\" 2>/dev/null",
        "description": "Find testnet directory and monitoring configs"
      }
    },
    {
      "tool": "shell_execute",
      "input": {
        "command": "ls -la /app/ 2>/dev/null | head -40",
        "description": "List app directory contents"
      }
    },
    {
      "tool": "shell_execute",
      "input": {
        "command": "find /app -maxdepth 3 -type d 2>/dev/null | head -60",
        "description": "Map app directory structure"
      }
    },
    {
      "tool": "shell_execute",
      "input": {
        "command": "ls -la /app/src/tools/ && cat /app/src/tools/prometheus.ts",
        "description": "Inspect prometheus tool"
      }
    },
    {
      "tool": "shell_execute",
      "input": {
        "command": "ls -la /app/src/metrics/ && find /app/src/metrics -type f",
        "description": "Inspect metrics directory"
      }
    }
  ],
  {
    "tool": "shell_execute",
    "input": {
      "command": "ls -la /app/src/metrics/ && find /app/src/metrics -type f",
      "description": "Inspect metrics directory"
    }
  }
],
{
  "tool": "shell_execute",
  "input": {
    "command": "ls -la /app/src/metrics/ && find /app/src/metrics -type f",
    "description": "Inspect metrics directory"
  }
}
```

```
"tool": "shell_execute",
"input": {
  "command": "cat /app/src/tools/prometheus.ts",
  "description": "Read prometheus tool source"
}
},
{
  "tool": "mongodb_list_collections",
  "input": {
    "includeStats": true
  }
},
{
  "tool": "shell_execute",
  "input": {
    "command": "which btcli 2>/dev/null; which python3 2>/dev/null; which docker 2>/dev/null",
    "description": "Check available tools"
  }
},
{
  "tool": "shell_execute",
  "input": {
    "command": "pip3 list 2>/dev/null | grep -i -E \"bittensor|substrate|prometheus|grafana\"",
    "description": "Check installed Python packages"
  }
},
{
  "tool": "shell_execute",
  "input": {
    "command": "which python3 || which python; which curl; which docker; which btcli; which",
    "description": "Find available executables"
  }
},
{
  "tool": "shell_execute",
  "input": {
    "command": "cat /app/src/metrics/registry.ts",
    "description": "Read metrics registry"
  }
},
{
  "tool": "mongodb_find_one",
  "input": {
    "collection": "deployer_state",
    "filter": {
      "procedure": "insignia_subnet_tuner"
    }
  }
},
{
  "tool": "mongodb_find_one",
```

```
"input": {
  "collection": "deployer_state"
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "defense_strategies",
    "filter": {
      "procedure": "insignia_subnet_tuner",
      "defense_id": "PC-VH-006"
    },
    "limit": 3
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "agent_memory",
    "filter": {
      "procedure": "insignia_subnet_tuner",
      "key": {
        "$regex": "knee_point|prometheus|monitoring|wallet|chain"
      }
    },
    "limit": 10
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "defense_strategies",
    "limit": 5,
    "projection": {
      "defense_id": 1,
      "status": 1,
      "name": 1
    }
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "agent_memory",
    "filter": {
      "key": {
        "$regex": "knee_point"
      }
    },
    "limit": 5
  }
}
```

```
},
{
  "tool": "shell_execute",
  "input": {
    "command": "find /app -name \"knee_point*\" -o -name \"state\" -type d 2>/dev/null | head -n 1",
    "description": "Find knee point config and state directory"
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "defense_strategies",
    "detail_level": "L2",
    "limit": 10,
    "projection": {
      "defense_id": 1,
      "status": 1,
      "name": 1,
      "description": 1,
      "type": 1
    }
  }
},
{
  "tool": "shell_execute",
  "input": {
    "command": "apt-get list --installed 2>/dev/null | grep -i -E \"curl|wget|docker|python3|python|python2\"",
    "description": "Check installed system packages"
  }
},
{
  "tool": "shell_execute",
  "input": {
    "command": "node -e \"try { const ws = new (require('ws'))('ws://127.0.0.1:9933'); ws.on('open', () => console.log('WebSocket connected')); } catch (e) { console.log('WebSocket connection failed'); }\"",
    "description": "Test WebSocket chain connectivity"
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "defense_strategies",
    "detail_level": "L2",
    "filter": {
      "$or": [
        {
          "defense_id": {
            "$regex": "PC-VH-006"
          }
        }
      ]
    },
    "name": {
```

```

        "$regex": "PC-VH-006"
      }
    },
    {
      "name": {
        "$regex": "Symbol Diversity"
      }
    },
    {
      "name": {
        "$regex": "Sybil"
      }
    },
    {
      "defense_id": {
        "$regex": "sybil"
      }
    }
  ]
},
"limit": 5
}
},
{
  "tool": "shell_execute",
  "input": {
    "command": "which docker 2>/dev/null; docker --version 2>/dev/null; docker ps 2>/dev/nul
    "description": "Check Docker availability"
  }
},
{
  "tool": "shell_execute",
  "input": {
    "command": "find / -name \"docker-compose*\" -type f 2>/dev/null | head -10; find / -nam
    "description": "Find subtensor and btcli binaries"
  }
},
{
  "tool": "shell_execute",
  "input": {
    "command": "mkdir -p /app/infrastructure/{scripts,config,monitoring,wallets,chain} 2>/de
    "description": "Create infrastructure directories and check prom-client"
  }
},
{
  "tool": "shell_execute",
  "input": {
    "command": "cat /app/package.json",
    "description": "Check package.json for dependencies"
  }
},

```

```

{
  "tool": "shell_execute",
  "input": {
    "command": "cat > /app/infrastructure/chain/connectivity_check.mjs << 'SCRIPT'\n/**\n *
    "description": "Create chain connectivity check script"
  }
},
{
  "tool": "shell_execute",
  "input": {
    "command": "node /app/infrastructure/chain/connectivity_check.mjs 2>&1",

...(truncated)

```

Errors (1)

[unknown] Untitled task

Error after 4 attempts: Cannot read properties of null (reading 'length')

Structured details:

```json

```

{
 "attempts": 4
}

```

Latest iteration output:

Error (final): Cannot read properties of null (reading 'length')

## Insights (1)

- **[orchestrator] Emulator Spec Execution State:** {"session": "EMULATOR\_SPEC\_execution\_v2", "timestamp": "2026-07-17T00:00:00Z", "loop\_phase": "wave1\_dispatched", "wave1\_tasks": {"tuner\_gen0\_init": {"task\_id": "6a3f3768d163a13dc7ec9046", "assignee": "tuner", "status": "pending", "description": "Initialize NSGA-II v5 gen0 population seeded with V13-R3 knee point"}, "deployer\_infra": {"task\_id": "6a3f377cd163a13dc7ec9048", "assignee": "deployer", "status": "pending", "description": "Verify chain, fund wallets, deploy monitoring, deploy PC-VH-006"}}

Task Details

| Status | Agent        | Title                                                                                   | Result                                                                                                  |
|--------|--------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Done   | orchestrator | Execute skill: EMULATOR_SPEC                                                            | Still pending. Let me give the agents more time to pick up and process these tasks. --<br>- *Used 58 to |
| Failed | unknown      | Untitled task                                                                           | Error after 4 attempts: Cannot read properties of null (reading 'length')                               |
| Done   | tuner        | Tuner: Initialize NSGA-II v5 gen0 population seeded with V13-R3 knee point              | The Gen0 population has been generated successfully. Now I'll store the population in MongoDB and up    |
| Done   | deployer     | Deployer: Verify chain, fund wallets, deploy monitoring stack, deploy PC-VH-006 defense | Good - the metrics endpoint is running but most values are 0 (not yet populated with current state).    |

Tool Usage

| Tool                     | Calls |
|--------------------------|-------|
| shell_execute            | 49    |
| read_task_result         | 34    |
| mongodb_find             | 21    |
| mongodb_aggregate        | 8     |
| read_memory              | 6     |
| mongodb_find_one         | 4     |
| mongodb_list_collections | 3     |
| write_task               | 3     |
| list_tasks               | 2     |
| dashboard_skill_execute  | 1     |
| list_memory_keys         | 1     |
| write_memory             | 1     |

Errors: 2

Agent Memory

Emulator Spec Execution State

*Written by orchestrator*

```
{"session": "EMULATOR_SPEC_execution_v2", "timestamp": "2026-07-17T00:00:00Z", "loop_phase":
"wave1_dispatched", "wave1_tasks": {"tuner_gen0_init": {"task_id": "6a3f3768d163a13dc7ec9046",
"assignee": "tuner", "status": "pending", "description": "Initialize NSGA-II v5 gen0 population seeded with
V13-R3 knee point"}, "deployer_infra": {"task_id": "6a3f377cd163a13dc7ec9048", "assignee": "deployer",
"status": "pending", "description": "Verify chain, fund wallets, deploy monitoring, deploy PC-VH-006"}},
"wave2_plan": {"simulator": "Run first online evaluation epoch with V13-R3 knee point config + full 14-
agent population — depends on deployer confirming chain ready"}, "wave3_plan": {"sentinel": "Evaluate
19 attack vectors, compute breach rate, check $9 acceptance gates — depends on simulator
completing"}, "system_state_snapshot": {"convergence": "gen0, ready_to_begin", "sentinel":
"SECURE_AND_IMPROVING, breach=0.0005, 18/19 vectors", "tuner": "clean_baseline_initialized, NSGA-II
v5, 41-dim, pop=30", "deployer": "OPERATIONAL_SIMULATION_READY, chain unconfirmed, wallets
unfunded", "prior_best_offline": "V13-R3-KP-020-a3c7: honest=0.9808, breach=2.6e-6, separation=0.963,
variance=0.00081, CR_eff=0.70", "acceptance_gates_status": "all_met_offline, needs_online_validation"},
"next_action": "poll_wave1_tasks"}
```

---

*Report generated by Insignia Swarm Gateway*